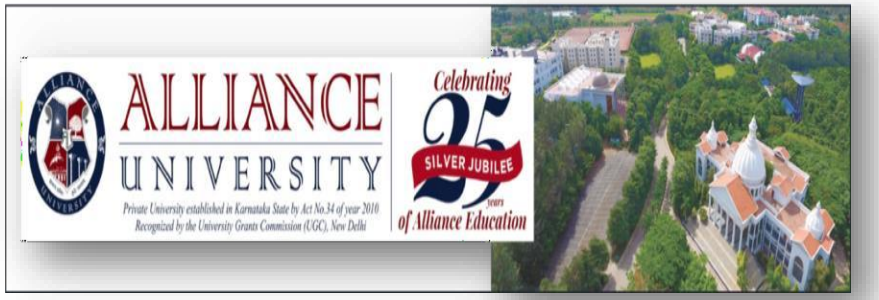
Crowd Management & Emergency System



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5IT1990 Design Project -1

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Introduction

- Enhances safety and emergency response at large-scale events like concerts, sports matches, and festivals.
- Integrates wearable health bands to continuously monitor heart rate and SpO_2 levels of attendees.
- Enables early detection of abnormal health conditions and triggers real-time alerts in emergencies.
- Provides precise location tracking for rapid medical assistance.
- Offers real-time crowd density analysis and visual dashboards for organizers.
- Combines AI analytics with IoT-based health tracking to create a smart, responsive, and health-aware system.

Review of literature

S.No	Author	Research/Methodology	Performance Metrics	Advantages	Disadvantages
1	Smith et al.	IoT-based crowd density estimation	Accuracy of density detection	Real-time monitoring	High power consumption
2	Lee et al.	AR-based indoor navigation	User navigation efficiency	Improved user experience	Requires high-end smartphones
3	Kumar et al.	Wearable health monitoring with ML	Anomaly detection accuracy	Proactive health alerts	Limited battery life
4	Johnson et al.	Emergency evacuation using ML	Evacuation time reduction	Optimized evacuation routes	Expensive hardware

Observations from the Literature Review

1. Most existing systems address crowd management and health monitoring as separate solutions.
2. Wearable health monitoring enables real-time data collection but faces battery and connectivity challenges.
3. AI models provide accurate anomaly detection but demand high computational power.
4. There is a lack of integrated systems combining real-time health tracking, crowd density visualization, and intelligent navigation in one unified platform.

Limitations from the Literature Review

1. Most studies focus on isolated aspects like crowd control or health monitoring, not an integrated system.
2. Wearable devices face battery drain and connectivity issues during continuous monitoring.
3. AI models for anomaly detection demand high computational power, affecting real-time performance.
4. There is a lack of real-time coordination between health data, crowd analytics, and emergency response mechanisms.

Problem Statement / Scope of the Project

- Managing large crowds and ensuring attendee safety is a major challenge during large-scale events.
- Existing systems focus separately on crowd management, navigation, or health monitoring.
- Lack of integration leads to delayed emergency response and inefficient coordination.
- The project aims to develop an AI-powered system combining wearable health tracking, real-time crowd analysis, and location-based emergency response.
- The solution ensures a safer, smarter, and more responsive event environment.

Proposed Idea / Solution

- Develop an AI-powered system that integrates wearable health bands to monitor attendees' heart rate and SpO_2 in real time.
- Use AI-based anomaly detection to identify abnormal health conditions automatically.
- Trigger instant emergency alerts and share location details with on-site medical teams for rapid response.
- Implement real-time crowd density visualization to prevent overcrowding and enhance situational awareness.
- Create a centralized dashboard for event organizers and medical staff to monitor overall safety and coordinate responses efficiently.

Objectives of the Idea / Solution as Design Project

- To monitor real-time heart rate and SpO_2 of attendees using wearable devices.
- To detect abnormal health conditions through AI-based anomaly detection.
- To trigger instant emergency alerts and provide accurate location tracking for quick medical response.
- To analyze crowd density for preventing overcrowding and improving safety management.
- To design a centralized monitoring dashboard for organizers to oversee health and safety conditions effectively.

Proposed Research / Methodology

- Use ESP32-based wearable bands with MAX30102 sensors to collect heart rate and SpO_2 data.
- Transmit data via BLE beacons to local gateways, then forward to cloud servers.
- Deploy AI models (TensorFlow Lite) to analyze vital signs and detect abnormal health conditions.
- Implement a backend system (FastAPI + Firebase/PostgreSQL) to store data, manage alerts, and visualize crowd density.
- Provide an event control center dashboard displaying real-time health monitoring, crowd density heatmaps, and emergency notifications.

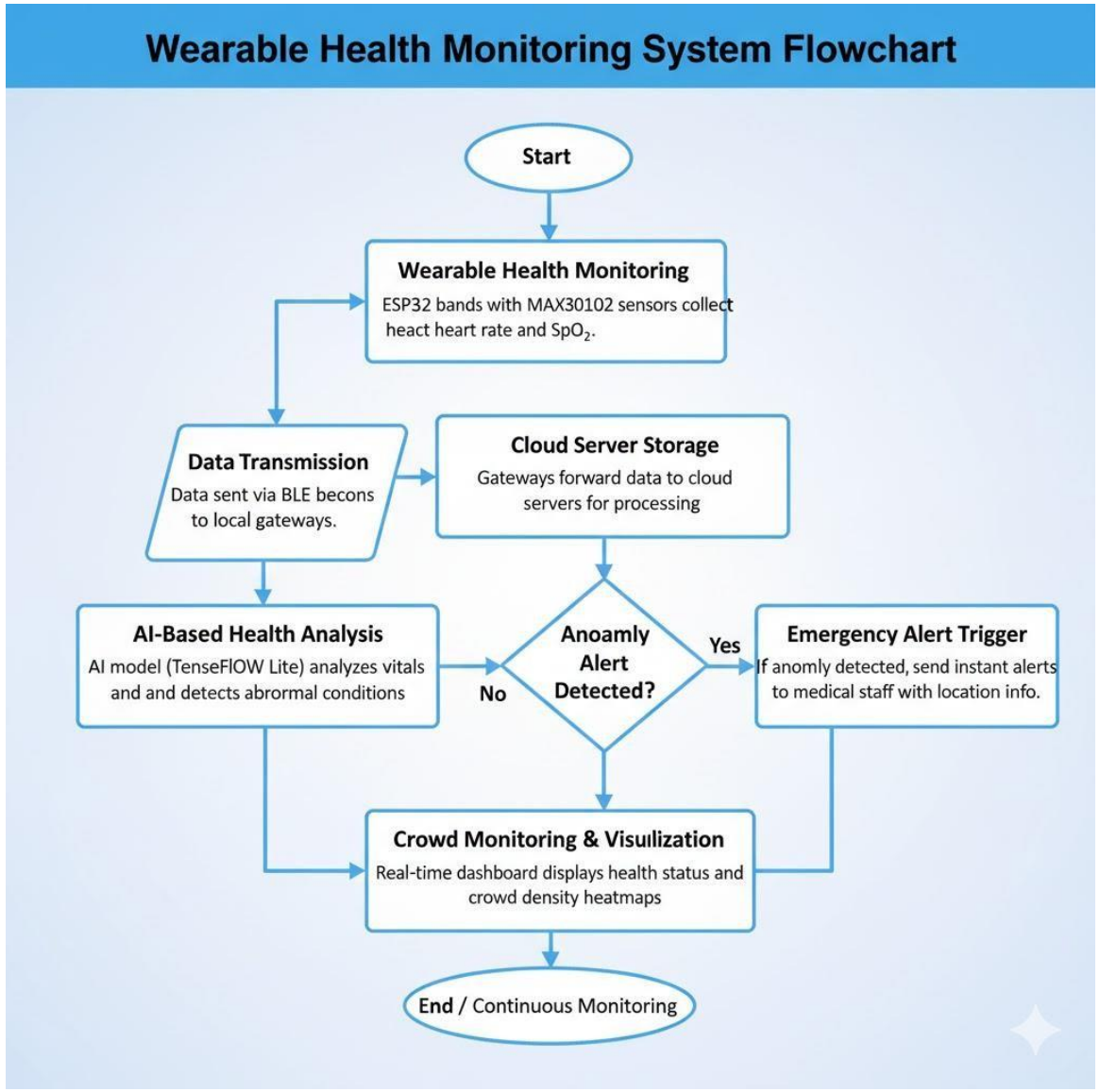
Hardware Requirements

- ESP32 microcontroller with MAX30102 sensors for heart rate and SpO_2 monitoring.
- BLE beacons for transmitting data and supporting crowd density tracking.
- Gateways for collecting data from wearables and sending it to cloud servers.
- Smartphones or tablets (optional) for viewing dashboards or alerts by organizers/medical staff.

Software Requirements

- TensorFlow Lite – for AI-based health anomaly detection.
- FastAPI – backend framework for data handling and alert management.
- Firebase / PostgreSQL – cloud database for storing health and crowd data.
- Dashboard/Visualization Tools – for real-time crowd density heatmaps and emergency notifications.
- Optional Mobile App – for organizers/medical staff to monitor health data and receive alerts.

Process / Flow / Design Diagram



Proposed Modules

1. Wearable Health Monitoring Module
   - ESP32 bands with MAX30102 sensors collect heart rate and SpO_2 data in real time.
2. Data Transmission Module
   - BLE beacons transmit data to local gateways, which forward it to cloud servers.
3. AI-Based Health Analysis Module
   - TensorFlow Lite models analyze vitals to detect abnormal conditions automatically.
4. Emergency Alert Module
   - Generates instant alerts with the location of affected individuals for rapid medical response.
5. Crowd Monitoring & Visualization Module
   - Displays real-time crowd density heatmaps and health status dashboards for organizers and medical teams.

Novelty of the Proposed Idea / Project

1. Integrated Approach: Combines wearable health monitoring, AI-based anomaly detection, and real-time crowd management in a single system.
2. Proactive Emergency Response: Automatically detects health issues and alerts medical teams instantly with precise location tracking.
3. Real-Time Crowd Insights: Provides live crowd density visualization, enabling organizers to prevent overcrowding and manage events efficiently.
4. IoT & AI Synergy: Leverages ESP32-based wearables and lightweight AI models for efficient, low-latency monitoring.
5. Scalable & Flexible: Can be deployed in concerts, sports events, festivals, or any large public gatherings.

Conclusion

1. The proposed AI-powered Crowd Management & Health Monitoring System ensures real-time monitoring of attendees' health during large-scale events.
2. It enables instant detection of abnormal conditions and triggers rapid emergency response with precise location tracking.
3. By combining wearable IoT devices, AI analytics, and crowd density visualization, the system enhances safety, efficiency, and situational awareness.
4. The project offers a scalable, integrated solution for safer and smarter management of public gatherings.

References

Authors, Year, Title of the Paper, Journal / Conference, Volume and Page Details, DOI (Samples are given below)

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3. Mooney, P. (2020). Kaggle chest x-ray images (pneumonia) dataset, https://github.com/ieee8023/covid-chestxray-datase

4. Parnian Afshar., Shahin Heidarian., Farnoosh Naderkhani., Anastasia Oikonomou., Konstantinos, N. Plataniotis., & Arash Mohammadi. (2020). COVID-CAPS: A capsule network-based framework for identification of COVID-19 cases from X-ray images, Pattern Recognition Letters, 138, 638–643

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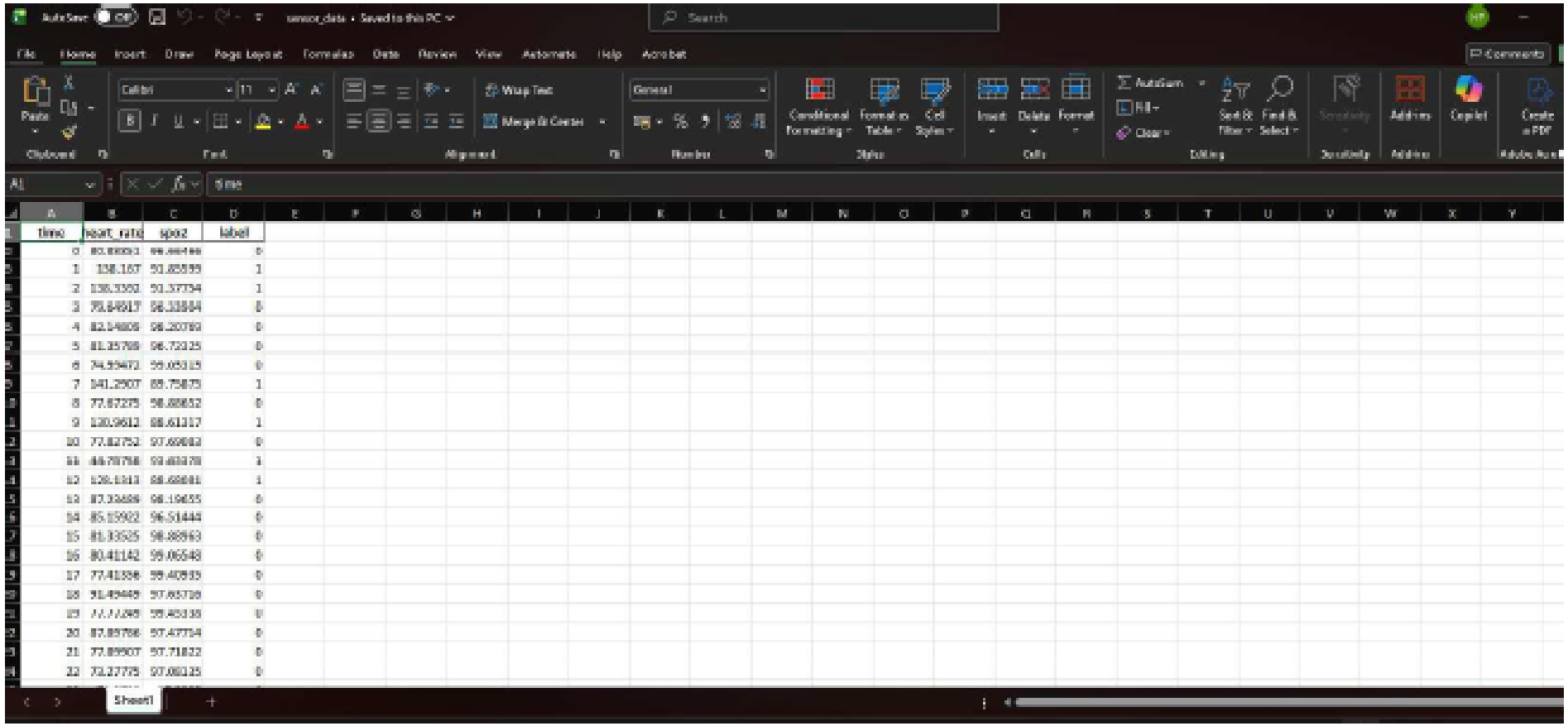
Partial Implementation

- Sample dataset to trail the model:



time	heart_rate	spo2	label
0	80.80091	99.46746	0
1	138.167	91.85595	1
2	138.3352	91.37754	1
3	95.84917	98.11884	0
4	82.14808	98.20782	0
5	81.35708	96.72125	0
6	74.55472	99.05315	0
7	141.2907	89.75875	1
8	77.87275	98.88852	0
9	120.9612	88.61317	1
10	77.82752	97.69882	0
11	44.78758	99.89378	1
12	128.1313	88.68881	1
13	87.32688	98.19655	0
14	85.15902	96.51444	0
15	81.33525	98.88963	0
16	80.41142	99.06548	0
17	77.41358	99.40835	0
18	91.49449	97.85726	0
19	77.77248	99.45138	0
20	87.85758	97.47754	0
21	77.89907	97.71822	0
22	72.27775	97.08125	0

Partial Implementation

- Training the XG Boost ML model using the sample data in python:

```
roj ML\main_ml.py'
  'Training model...
d:\PYTHON COMPILER\lib\site-packages\xgboost\core.py:158: UserWarning: [09:18:33] WARNING: C:\buildkite-agent\builds\bu
c:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)

   Accuracy: 0.964824120603015
              precision    recall  f1-score   support

           0       0.98      0.98      0.98       173
           1       0.88      0.85      0.86        26

    accuracy                           0.96       199
   macro avg       0.93      0.91      0.92       199
weighted avg       0.96      0.96      0.96       199


   Model saved as xgb_heart_model.pkl
```

Partial Implementation

- Testing the Trained XGBoost ML model using python:

```
PS E:\Design Proj ML>  e:; cd 'e:\Design Proj ML'; & 'd:\PYTHON COMPILER\python.exe' 'c:\Users\TONYHOLMES\.
roj ML\test_model.py'
   Model loaded successfully!

   Test Accuracy: 0.9919597989949749
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       836
           1       0.97      0.97      0.97       159

    accuracy                           0.99       995
   macro avg       0.99      0.99      0.99       995
weighted avg       0.99      0.99      0.99       995


   Predictions for new samples: [0 1 1]
PS E:\Design Proj ML>
```